RGPV • B.TECH • SEMESTER II

ENGINEERING PHYSICS

(BT-201)

UNIT 1

Quantum Mechanics

📘 TOPICS COVERED

1. *Heisenberg's Uncertainty Principle*
2. *Time Independent Schrödinger Equation*
3. *Phase Velocity & Group Velocity*
4. *Time Dependent Schrödinger Equation*
5. *Particle in a One-Dimensional Box*

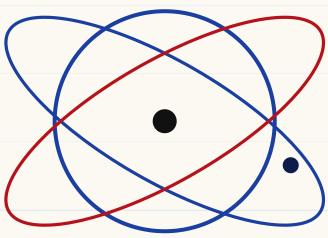

fig — electron orbiting nucleus (Bohr model, for reference)

Handwritten Revision Notes • Neatly Explained • Exam-Focused

TOPIC 1

Heisenberg's Uncertainty Principle

Background: Matter has dual nature — it behaves both as a particle and as a wave (de Broglie). Because of this wave nature, a moving particle cannot be treated like a tiny ball with an exact position and exact speed at the same time.

➤ Statement of the Principle

"It is impossible to measure simultaneously, with perfect accuracy, both the position and the momentum of a microscopic particle. There is always some uncertainty (error) in one or both."

If Δx = uncertainty in position, and Δp = uncertainty in momentum, then:

$$\Delta x \cdot \Delta p \geq h / 4\pi$$

h = Planck's constant = 6.626×10^{-34} J·s

In terms of $\hbar$ (h-bar = $h/2\pi$), this is also written as:

$$\Delta x \cdot \Delta p \geq \hbar / 2$$

➤ Other Uncertainty Relations (same idea, other pairs)

Conjugate Pair	Relation
Energy & Time	$\Delta E \cdot \Delta t \geq h/4\pi$
Angular momentum & Angle	$\Delta L \cdot \Delta\theta \geq h/4\pi$

⭐ **EXAM TIP: $\Delta E \cdot \Delta t \geq h/4\pi$ is used to explain the natural line-width / lifetime of excited atomic states — a favourite RGPV question!**

➤ Why does this happen? (Physical meaning)

It is NOT because our instruments are bad. It is a fundamental property of nature. The very act of measuring disturbs the particle. To "see" an electron we must bounce something (like a photon) off it — and that bounce itself changes the electron's momentum.

➤ Explanation by Gamma-Ray Microscope (Thought Experiment)

Suppose we want to "see" an electron using a super powerful gamma-ray microscope.

- *To locate the electron precisely (small Δx), we need light of a very short wavelength λ (gamma rays) — because resolving power improves as λ decreases.*
- *But short wavelength ⇒ high frequency ⇒ high photon energy & momentum ($E = hc/\lambda$).*
- *When this high-energy photon strikes the electron, it gives the electron a large, unpredictable "kick" — so the momentum of the electron becomes very uncertain (large Δp).*

Conclusion: The better we try to fix the position ($\Delta x \downarrow$), the worse the momentum uncertainty becomes ($\Delta p \uparrow$), and vice-versa. Their product can never become zero.

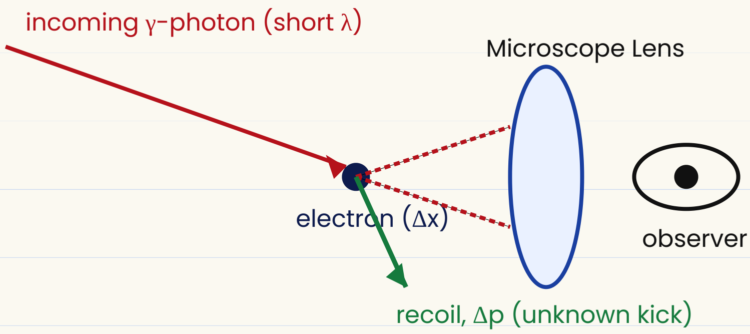

fig — Heisenberg's gamma-ray microscope thought experiment

> **TRICK TO REMEMBER:** *"The Sharper the Spot, the Wilder the Speed" — sharp position knowledge (small Δx) always comes at the cost of wild momentum uncertainty (large Δp).*

➤ Important Application: Non-existence of electron inside the nucleus

If an electron were confined inside a nucleus (size ≈ 10^{-14} m), $\Delta x \approx 10^{-14}$ m. Using $\Delta x \cdot \Delta p \geq h/4\pi$, the minimum momentum works out so large that the electron's calculated energy becomes far greater than the energy actually observed being emitted in beta decay (~MeV range mismatch, calculated ~20 MeV vs observed ~3-4 MeV). This proves electrons cannot exist inside the nucleus — a classic RGPV numerical question.

✔ QUICK REVISION

- *$\Delta x \cdot \Delta p \geq h/4\pi$ — cannot know position & momentum exactly together.*
- *Also applies to Energy–Time and Angular momentum–Angle.*
- *Explained using the gamma-ray microscope experiment.*
- *Used to prove electrons cannot exist inside the nucleus.*

TOPIC 2

Time Independent Schrödinger Equation (TISE)

A moving particle is associated with a "matter wave" (de Broglie wave). Schrödinger described this wave using a mathematical function called the wave function, **ψ** *(psi). The equation that ψ must satisfy is called the Schrödinger Wave Equation.*

➤ Derivation (Simple Steps)

Step 1 — *A particle of mass m moving with velocity v behaves as a wave. Its wave function (like a classical wave) can be written as:*

$$\psi(x,t) = \psi_0 \sin(2\pi/\lambda)(x - vt)$$

Step 2 — *Differentiate ψ twice with respect to x:*

$$d^2\psi/dx^2 = -(4\pi^2/\lambda^2)\psi$$

Step 3 — *Use de Broglie's relation $\lambda = h/mv$, so $1/\lambda^2 = m^2v^2/h^2$. Substitute:*

$$d^2\psi/dx^2 + (4\pi^2 m^2 v^2 / h^2)\psi = 0$$

Step 4 — *Total energy E = K.E. + P.E. = $\frac{1}{2}mv^2 + V$, so $m^2v^2 = 2m(E - V)$. Substituting this in place of m^2v^2:*

$$d^2\psi/dx^2 + (8\pi^2 m / h^2)(E - V)\psi = 0$$

↑ *This is the Time-Independent Schrödinger Equation in one dimension — the most important formula of this topic!*

➤ Operator Form

Writing it using the Hamiltonian operator $\hat{H}$:

$$\hat{H}\psi = E\psi$$

where $\hat{H} = -(\hbar^2/2m)(d^2/dx^2) + V(x)$ is the total energy (Hamiltonian) operator.

⭐ **EXAM TIP: Deriving this equation (Steps 1–4) is a very common 7-mark RGPV question. Practice writing every step.**

➤ Physical Significance of ψ and |ψ|²

- *ψ itself has no direct physical meaning — it can even be a complex/negative quantity.*

- $|\psi|^2$ ($= \psi \cdot \psi^*$) gives the probability density — i.e. the probability of finding the particle per unit volume at that point. This is called the Born interpretation.
- Probability of finding particle in small volume $d\tau = |\psi|^2 d\tau$.

➤ Conditions for a "Well-Behaved" (Acceptable) Wave Function

Condition	Meaning
Single-valued	only one value of ψ at each point x
Finite	ψ must not become infinite anywhere
Continuous	ψ and $d\psi/dx$ have no sudden jumps
Normalizable	$\int|\psi|^2 d\tau = 1$ over all space

TRICK: *Remember "SFCN" — Single valued, Finite, Continuous, Normalizable — to recall the 4 conditions instantly.*

✔ QUICK REVISION

- TISE: $d^2\psi/dx^2 + (8\pi^2 m/h^2)(E-V)\psi = 0$, derived from de Broglie wave + energy conservation.
- Operator form: $\hat{H}\psi = E\psi$
- $|\psi|^2$ = probability density (Born interpretation).
- ψ must be single-valued, finite, continuous, and normalizable.

TOPIC 3

Phase Velocity & Group Velocity

A single pure wave of exact wavelength would spread over all space — it cannot represent a localized (particle-like) object. So a real particle is represented by a wave packet / wave group — a superposition of many waves of slightly different wavelengths (and frequencies), which add up to a bump localized in a small region.

➤ (a) Phase Velocity (v_p)

The velocity with which a single wave (a point of constant phase, like a crest) travels.

$$\boxed{v_p = \omega / k = \nu \lambda}$$

where ω = angular frequency, $k = 2\pi/\lambda$ = wave number.

Using $E = h\nu$ and $p = h/\lambda \Rightarrow \omega = E/\hbar$ and $k = p/\hbar$, so:

$$\boxed{v_p = E / p}$$

➤ (b) Group Velocity (v_g)

The velocity with which the "envelope" of the wave packet (i.e. the resultant bump/group) travels. This is the velocity at which energy — and the particle itself — actually travels.

$$\boxed{v_g = d\omega / dk}$$

Important result — *it can be shown that the group velocity equals the actual particle velocity:*

$$\boxed{v_g = v \quad \text{(particle's actual velocity)}}$$

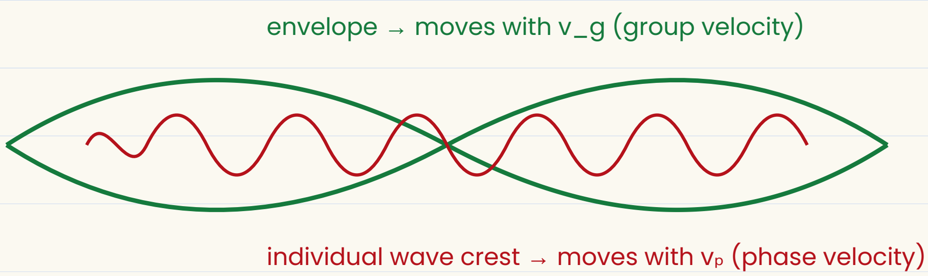

fig — a wave packet: carrier wave (red) inside envelope (green)

➤ Relation between v_p and v_g

$$\boxed{v_p \cdot v_g = c^2}$$

Since v_g (particle speed) is always less than c, this means v_p is always greater than c — but this does NOT violate relativity because v_p carries no energy/information, only v_g does.

⭐ **EXAM TIP: "Group velocity = particle velocity, Phase velocity can exceed c but carries no information" — this line is frequently asked as a short-answer question.**

TRICK: *"Group = Goods" — the Group velocity carries the "goods" (the particle, energy, information); phase velocity is just the wiggle inside.*

✔ QUICK REVISION

- $v_p = \omega/k = E/p$ — velocity of individual wave crest.
- $v_g = d\omega/dk$ = actual particle velocity.
- $v_p \cdot v_g = c^2 \rightarrow v_p$ can exceed c, v_g cannot.

TOPIC 4

Time Dependent Schrödinger Equation (TDSE)

The TISE (Topic 2) cannot describe how ψ changes with time — it only works for constant-energy (stationary) states. The TDSE is the more general, master equation of quantum mechanics that gives the full evolution of ψ(x,t).

➤ Derivation (Simple Steps)

Step 1 — *Take the plane wave function of a free particle:*

$$\psi = \psi_0\, e\text{\^{}}\{-i(2\pi/h)(Et - px)\}$$

Step 2 — *Differentiate ψ once with respect to time t:*

$$\partial\psi/\partial t = -(2\pi i E/h)\psi \;\Rightarrow\; E\psi = (ih/2\pi)\,\partial\psi/\partial t$$

Step 3 — *From TISE we know* $E\psi = -(h^2/8\pi^2 m)(d^2\psi/dx^2) + V\psi$. *Equating the two expressions for Eψ:*

$$ih/2\pi\ \partial\psi/\partial t = -(h^2/8\pi^2 m)\,\partial^2\psi/\partial x^2 + V\psi$$

Using $\hbar = h/2\pi$, *the compact standard form is:*

$$i\hbar\ \partial\psi/\partial t = \hat{H}\psi = -(\hbar^2/2m)\,\partial^2\psi/\partial x^2 + V\psi$$

↑ *This is the Time-Dependent Schrödinger Equation — considered the fundamental equation of quantum mechanics.*

⭐ **EXAM TIP: RGPV often asks — "Derive TDSE from TISE" and "distinguish between TISE and TDSE" (table below is perfect for that answer).**

➤ TISE vs TDSE — Quick Comparison

Time-Independent (TISE)	Time-Dependent (TDSE)
$d^2\psi/dx^2 + (8\pi^2 m/h^2)(E-V)\psi = 0$	$i\hbar\,\partial\psi/\partial t = \hat{H}\psi$
ψ depends only on position x	*ψ depends on position x and time t*
Used for stationary (steady) states	*Used for any general state, incl. changing systems*

TRICK: *TDSE has an "i" (iota) and "∂/∂t" in it — if you see i and t together in a Schrödinger-looking equation, it's the Time-Dependent one!*

✔ QUICK REVISION

- *TDSE: $i\hbar\, \partial\psi/\partial t = \hat{H}\psi$, derived from the plane wave + TISE.*
- *Describes how $\psi(x,t)$ evolves with both space and time.*
- *Reduces to TISE when the system is in a stationary (constant energy) state.*

TOPIC 5

Particle in a One-Dimensional Box

This is the simplest quantum-mechanical problem, showing how confining a particle automatically leads to quantized energy. Consider a particle of mass m free to move only along the x-axis between $x = 0$ and $x = L$, trapped by infinitely high walls at both ends (an "infinite potential well").

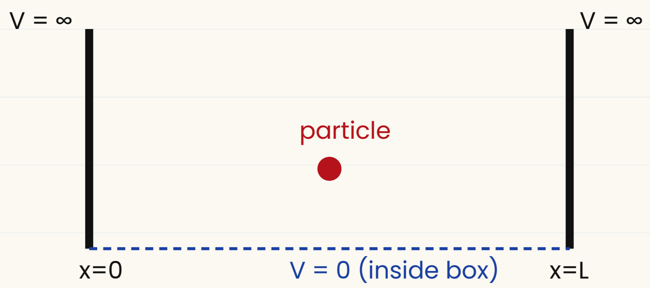

fig — infinite potential well (box) of width L

➤ Boundary Conditions

Inside the box $V = 0$, so TISE becomes: $d^2\psi/dx^2 + (8\pi^2mE/h^2)\psi = 0$, i.e. $d^2\psi/dx^2 + k^2\psi = 0$ where $k^2 = 8\pi^2mE/h^2$.

The general solution is: $\psi(x) = A\sin(kx) + B\cos(kx)$

- *Since the walls are infinitely high, particle can NEVER be found outside ⇒ $\psi = 0$ at $x=0$ and $x=L$.*
- *At $x = 0$: $\psi(0)=0 \Rightarrow A\sin 0 + B\cos 0 = 0 \Rightarrow B = 0$.*
- *At $x = L$: $\psi(L)=0 \Rightarrow A\sin(kL) = 0 \Rightarrow kL = n\pi \quad (n = 1,2,3,\dots)$*

So $k = n\pi/L$. ($n = 0$ is rejected — it would mean $\psi = 0$ everywhere, i.e. no particle at all.)

➤ Quantized Energy Levels

Putting $k = n\pi/L$ back into $k^2 = 8\pi^2mE/h^2$ and solving for E:

$$\boxed{E_n = n^2h^2/8mL^2 \quad (n = 1, 2, 3, \dots)}$$

The energy is quantized — the particle can only have these specific, discrete energy values, not just any value. This is the most important formula of the topic.

⭐ EXAM TIP: "Derive expression for energy of a particle in a 1-D box" is almost guaranteed every year in RGPV BT-201.

➤ Normalized Wave Function

Applying the normalization condition $\int_0^L |\psi|^2 dx = 1$ gives $A = \sqrt{(2/L)}$. So the complete wave function is:

$$\boxed{\psi_n(x) = \sqrt{(2/L)}\ \sin(n\pi x / L)}$$

➤ Zero-Point Energy

The lowest possible energy is at $n = 1$ (ground state) — NOT zero!

$$\boxed{E_1 = h^2 / 8mL^2 \ \ \text{(zero-point energy, minimum possible energy)}}$$

This shows that a confined quantum particle can never be perfectly at rest — it always has some minimum energy, consistent with the uncertainty principle (Δx is finite, so $\Delta p \neq 0$).

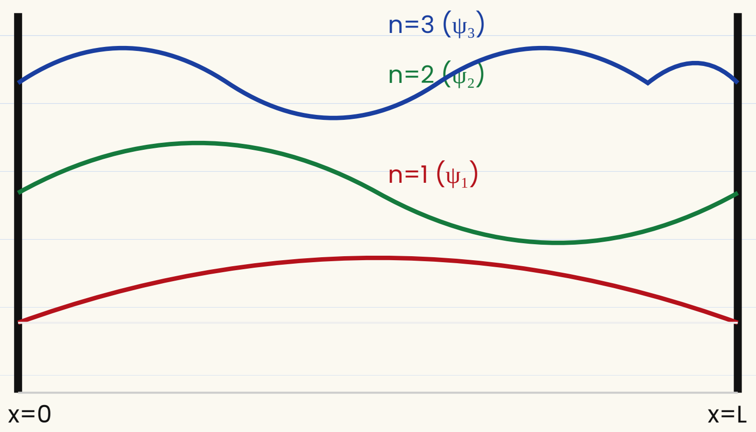

fig — wave functions ψ_n for n = 1, 2, 3 inside the box

➤ Energy Level Diagram

n	Energy E_n	Nodes (points where ψ=0, excl. ends)
1	$1 \times h^2/8mL^2$	*0*
2	$4 \times h^2/8mL^2$	*1*
3	$9 \times h^2/8mL^2$	*2*
n	$n^2 \times h^2/8mL^2$	$(n-1)$

TRICK: *"Energy goes as n-square" — $E_1 : E_2 : E_3 = 1:4:9$. Number of nodes (inside box) = $n - 1$.*

✔ QUICK REVISION

- *Boundary conditions $\psi(0)=\psi(L)=0$ force $k = n\pi/L$ → quantized energy.*
- $E_n = n^2h^2 / 8mL^2$, $\quad \psi_n(x) = \sqrt{(2/L)}\sin(n\pi x/L)$
- *Minimum energy is NOT zero → Zero-point energy $E_1 = h^2/8mL^2$.*

- *Number of nodes inside the box* $= n - 1$.

Unit 1 — Formula Sheet (Last-Minute Revision)

Topic	Key Formula
Heisenberg's Uncertainty Principle	$\Delta x \cdot \Delta p \geq h/4\pi$; $\Delta E \cdot \Delta t \geq h/4\pi$
Time Independent Schrödinger Eqn	$d^2\psi/dx^2 + (8\pi^2 m/h^2)(E-V)\psi = 0$
Phase velocity	$v_p = \omega/k = E/p$
Group velocity	$v_g = d\omega/dk$ = particle velocity ; $v_p \cdot v_g = c^2$
Time Dependent Schrödinger Eqn	$i\hbar\, \partial\psi/\partial t = -(\hbar^2/2m)\partial^2\psi/\partial x^2 + V\psi$
Particle in a box — Energy	$E_n = n^2h^2 / 8mL^2$
Particle in a box — Wave fn	$\psi_n(x) = \sqrt{(2/L)}\, \sin(n\pi x/L)$

➤ Frequently Asked (Previous-Year Style) Questions

- *State and explain Heisenberg's Uncertainty Principle. Prove that electron cannot exist inside the nucleus.*
- *Derive the Time-Independent Schrödinger wave equation.*
- *What are the physical significance and properties of a well-behaved wave function?*
- *Define phase velocity and group velocity. Derive the relation* $v_p \cdot v_g = c^2$.
- *Derive the Time-Dependent Schrödinger equation.*
- *Set up the Schrödinger equation for a particle in a 1-D box and obtain its energy eigenvalues and eigenfunctions.*
- *Show that the energy of a particle in a box is quantized, and explain the zero-point energy.*

⭐ **OVERALL EXAM TIP: Numericals from Heisenberg's principle and Particle-in-a-box (finding E_n, Δx, Δp for given L or m) are asked almost every year — practice at least 5 numericals from each.**

MASTER MNEMONIC — "H-T-P-T-P": *Heisenberg → TISE → Phase/Group velocity → TDSE → Particle in a box — the natural flow of Unit 1, in the order it is usually taught & asked.*

✔ UNIT 1 IN ONE LINE

Quantum Mechanics tells us matter behaves as a wave (ψ), we can never know position & momentum together exactly (Heisenberg), ψ obeys the Schrödinger equations (time-independent & time-dependent), and confining a particle (like in a box) automatically quantizes its energy.